

Particle in a One-Dimensional Box

Energy Quantisation, Probability Density and Heisenberg Uncertainty

BPhO Computational Challenge 2026 — Task 7

Abstract

This report implements the exact stationary states of a non-relativistic particle confined by an infinite one-dimensional potential well. The energy spectrum and Born probability densities satisfy the official Task 7 requirements. The extension evaluates the position and momentum moments and shows analytically and computationally that every state obeys $\Delta x \Delta p \geq \hbar/2$. A finite-difference eigensolver provides an independent numerical check of the analytical solution.

1 Physical model

The potential is zero inside a box of width a and infinite outside:

$$V(x) = \begin{cases} 0, & 0 < x < a, \\ \infty, & x \leq 0 \text{ or } x \geq a. \end{cases} \quad (1)$$

The wavefunction must therefore vanish at both walls, $\psi(0) = \psi(a) = 0$. Within the box the time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi. \quad (2)$$

The boundary conditions select positive integer quantum numbers $n = 1, 2, \dots$ and normalized eigenstates

$$\psi_n(x, t) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{-iE_n t/\hbar}. \quad (3)$$

The corresponding energy spectrum is

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2} = \frac{n^2 h^2}{8ma^2}. \quad (4)$$

The computational baseline uses an electron and $a = 1.00 \text{ nm}$.

2 Probability density

The Born probability density is

$$|\psi_n(x, t)|^2 = \frac{2}{a} \sin^2\left(\frac{n\pi x}{a}\right). \quad (5)$$

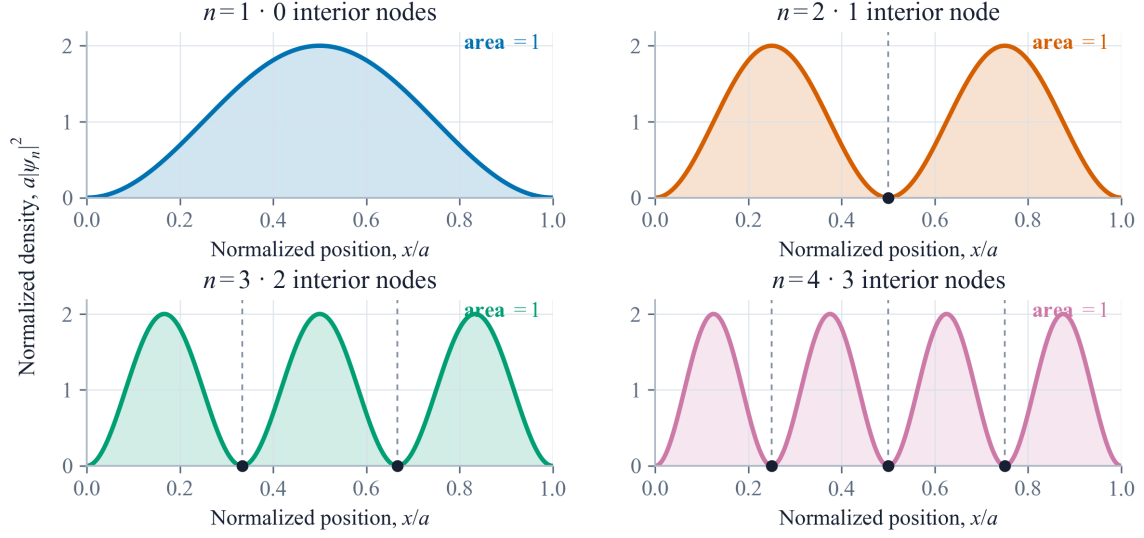
The factor $e^{-iE_n t/\hbar}$ has unit modulus, so a single energy eigenstate has a time-independent probability density. Normalisation follows from

$$\int_0^a |\psi_n|^2 dx = \frac{2}{a} \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = 1. \quad (6)$$

State n has $n - 1$ interior nodes at $x = ka/n$, where $k = 1, \dots, n - 1$.

First four stationary-state probability densities

$a|\psi_n|^2 = 2\sin^2(n\pi x/a)$ · each curve integrates to one and vanishes at both walls



Higher-energy states have more nodes, but every density remains symmetric about the centre of the box.

Figure 1: Dimensionless probability densities for the first four stationary states. The area under every curve is one.

3 Position uncertainty

The expectation value of position is

$$\langle x \rangle = \int_0^a x |\psi_n|^2 dx = \frac{a}{2}. \quad (7)$$

The second moment is

$$\langle x^2 \rangle = \frac{2}{a} \int_0^a x^2 \sin^2\left(\frac{n\pi x}{a}\right) dx \quad (8)$$

$$= a^2 \left(\frac{1}{3} - \frac{1}{2n^2\pi^2} \right). \quad (9)$$

Consequently,

$$(\Delta x)^2 = \langle x^2 \rangle - \langle x \rangle^2 \quad (10)$$

$$= a^2 \left(\frac{1}{12} - \frac{1}{2n^2\pi^2} \right), \quad (11)$$

and

$$\Delta x = a \sqrt{\frac{1}{12} - \frac{1}{2n^2\pi^2}}. \quad (12)$$

4 Momentum uncertainty

The momentum operator is

$$\hat{p} = -i\hbar \frac{d}{dx}. \quad (13)$$

Because each spatial eigenfunction is real and vanishes at the boundaries,

$$\langle p \rangle = \int_0^a \psi_n^* \left(-i\hbar \frac{d}{dx} \right) \psi_n dx \quad (14)$$

$$= -\frac{i\hbar}{2} [\psi_n^2]_0^a = 0. \quad (15)$$

Applying the momentum operator twice gives

$$\langle p^2 \rangle = \int_0^a \psi_n^* \left(-\hbar^2 \frac{d^2}{dx^2} \right) \psi_n dx \quad (16)$$

$$= \frac{n^2 \pi^2 \hbar^2}{a^2}. \quad (17)$$

Thus

$$\Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \frac{n\pi\hbar}{a}. \quad (18)$$

5 Verification of the uncertainty principle

Multiplying Eqs. (12) and (18) gives

$$\Delta x \Delta p = \hbar \sqrt{\frac{n^2 \pi^2}{12} - \frac{1}{2}}. \quad (19)$$

The expression increases with positive integer n , so its minimum occurs at $n = 1$:

$$\frac{\Delta x \Delta p}{\hbar} = \sqrt{\frac{\pi^2}{12} - \frac{1}{2}} = 0.567861808 \dots > \frac{1}{2}. \quad (20)$$

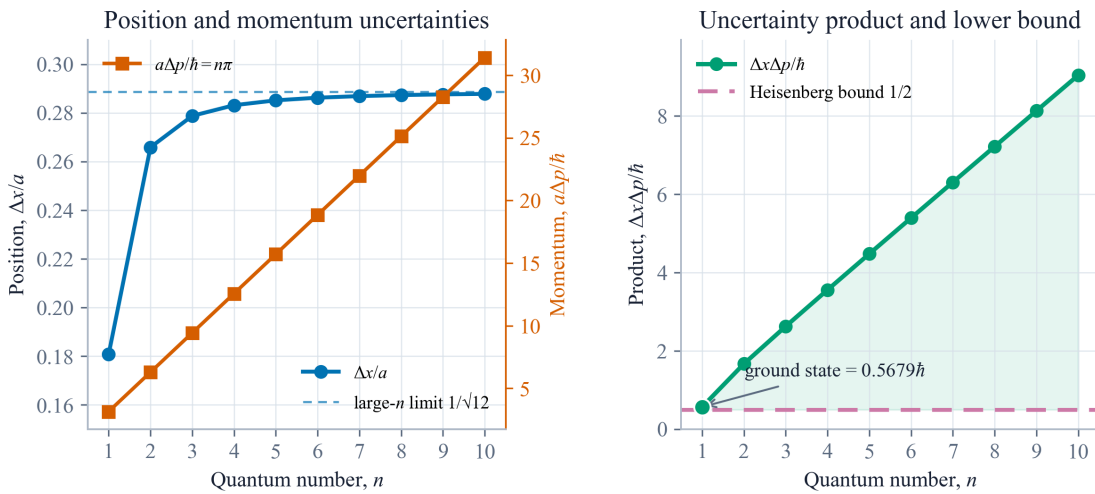
Therefore every stationary state in the box satisfies Heisenberg's relation

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \quad (21)$$

The box ground state is not a minimum-uncertainty Gaussian state, so the inequality is strict.

Heisenberg uncertainty in the particle-in-a-box

$$\Delta x \Delta p / \hbar = \sqrt{(n^2 \pi^2 / 12 - 1/2)} \geq 1/2$$



The ground state gives the smallest product; momentum uncertainty then grows with quantum number.

Figure 2: Position and momentum uncertainties and their product for $n = 1, \dots, 10$. The dashed magenta line marks the Heisenberg bound.

6 Computational implementation and validation

The analytical model evaluates Eqs. (4)–(19) on a 2001-point position grid. A separate finite-difference model approximates the second derivative by

$$\left. \frac{d^2\psi}{dx^2} \right|_j \approx \frac{\psi_{j+1} - 2\psi_j + \psi_{j-1}}{(\Delta x)^2}, \quad (22)$$

creating a symmetric tridiagonal Hamiltonian. Its eigenvalues are calculated without inserting the analytical spectrum. Position and momentum moments are then computed directly from each normalized numerical eigenvector using

$$\langle A \rangle_h = \sum_j \psi_j^* (A_h \psi)_j \Delta x, \quad (p_h \psi)_j = -i\hbar \frac{\psi_{j+1} - \psi_{j-1}}{2\Delta x}. \quad (23)$$

The same zero-valued boundary points used by the Hamiltonian close the first- and second-derivative stencils. Analytical uncertainties are introduced only after the numerical $\langle x \rangle$, $\langle x^2 \rangle$, $\langle p \rangle$ and $\langle p^2 \rangle$ have been evaluated.

Table 1: Maximum relative energy error over the first ten numerical states.

Interior points	Maximum relative error
100	8.04×10^{-3}
200	2.03×10^{-3}
400	5.11×10^{-4}
800	1.28×10^{-4}
1600	3.21×10^{-5}

For $N = 1600$, the maximum relative errors in Δp and $\Delta x \Delta p$ are both below 1.605×10^{-5} . Their measured convergence orders over all ten states lie between 1.99961 and 2.00038, consistent with second-order discretisation. The energy orders lie between 1.99896 and 1.99996. The finest-grid absolute overlap between every numerical eigenvector and its analytical counterpart is greater than 0.999999. All fifty grid-state uncertainty products satisfy $\Delta x \Delta p \geq \hbar/2$. The 50-check acceptance suite additionally checks normalisation, orthogonality, node counts, boundary values, energy scaling, expectation values and the uncertainty inequality.

7 Interpretation and limitations

The infinite well is an ideal model: real confining potentials are finite and may allow tunnelling. The calculation is one-dimensional, non-relativistic and contains one non-interacting particle with no external field or spin dynamics. These assumptions do not affect the internal verification of the specified Task 7 model, but they limit direct quantitative comparison with a particular laboratory system.

Reproducibility. The source code, CSV evidence, reference anchors, validation report, PNG/SVG figures and deterministic manifest are generated by a single command. The official brief is available from the British Physics Olympiad Computational Challenge page.